

MERGER AND SCATTERING OF TRAVERSABLE WORMHOLES: GRAVITATIONAL RADIATION FROM UNSTABLE BINARIES

NIKITA M. SHIROKOV*

Independent Researcher, Moscow, Russia

(Dated: September 19, 2026)

We evolve pairs of massive drainhole wormholes—phantom-scalar throats—in full numerical relativity and map every outcome a pair can have. A lone throat is an unstable fixed point (e-fold time $\tau = 5.9$, in the González–Guzmán–Sarbach band) with two branches, collapse and inflation; a declared kick selects the branch deterministically and accelerates it. Like-oriented throats repel, so a merger requires opposite scalar signs. A head-on pair then collides with both mouths still open and forms a black hole, the trapped surface born enclosing both mouths and never either alone, which loses mass to the phantom field it swallows ($M_{\text{MS}} 3.0 \rightarrow 2.2$) and rings without a chirp. The spiral merger instead stalls: no horizon exists anywhere on its measured record, the curvature wall that kills every grid rides the still-open throat, and a $16\times$ refinement ladder saturates without reaching the merger’s end. An e-fold budget built on three agreeing measurements of the unstable mode forbids a wormhole inspiral outright—an orbit costs $(d/M)^4$ while a throat endures only $\ln(1/\varepsilon)$, so surviving one orbit from the innermost stable circular orbit needs a seed below 10^{-20} . Every encounter is therefore a plunge, and the loudest channel is the horizonless fly-by, radiating $7.4 \times 10^{-2} M$ in the dominant multipole— $29\times$ an equal-parameter vacuum binary. Rescaled to stellar mass and carried through the detector data products, these bursts are not separated from binary black holes by morphology at fixed duration (fitting factors 0.82–0.98 against a vacuum control’s 0.94–0.95); the discriminator is the frequency track, which falls or drifts where a remnant’s would hold at the Kerr value. A matched-filter search of public Advanced LIGO O3b strain with a 127-template bank returns no candidate at zero lag, at a measured single-detector reach of 0.15–4.8 Gpc across the channels.

I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale, both mouths in the same space [1–3]. More than one mechanism promotes them to macroscopic size: a Planck-scale Lorentzian wormhole caught in an inflating background is stretched with it and emerges macroscopic [4]; bubble and domain-wall cosmologies produce, without tuning, post-inflationary regions joined through wormholes [5, 6]; and a gas of relic wormholes has been run as a cosmology in its own right [7, 8]. What none of this work evolves is the aftermath: what a relic wormhole does once it is macroscopic and classical, and what two of them do to each other. That aftermath is this paper’s subject.

The classical theory fixes its two clocks. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable: perturbed one way the throat collapses to a black hole, the other way it inflates, on a timescale set by the throat radius [13, 14]. Every constituent destroys itself on its own schedule. Second, Geroch’s theorem forbids classical topology change [15, 16]: two wormholes cannot fuse into one, and none can be born. Birth is quantum [17], so a simulation takes over at $t = 0$ with the newborn state as its initial data—the stipulation every numerical-relativity binary makes, here with a physical story attached.

Together the two facts turn the standing objection to wormhole binaries into a selection rule. A binary has no astrophysical assembly channel—each throat dies faster than capture could pair two—but a foam-born pair needs none: creation and collision are the same event, at a birth separation of order the objects themselves. The race is collision against decay, and both rates belong to the classical solution. The scalar’s sign adds a second rule at no cost: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard routes [18–20]—or inflation. A foam population thus fans out into four fates—merge, scatter, collapse, inflate—across the three-parameter space of birth separation, perturbation and relative sign, and that space is what a family of evolutions maps. The binary evolved here is the exterior approximation of a pair of handles (three asymptotic ends, the far sides passive); what merges is not two wormholes into one, which Geroch forbids, but two wormholes into one black hole with both throats inside it—the horizon does the work that topology change cannot.

The observable is not a chirp but a stochastic background from the merger epoch, redshifted to whichever band the mass scale puts it in [21–24]. The clean deliverable is therefore dimensionless and epoch-independent: the energy radiated per merger and its spectral shape in throat-mass units, which converts any bound on Ω_{GW} —including the nucleosynthesis cap that holds outside every detector band [25]—into a bound on the co-moving abundance of primordial wormhole mergers. Two

* shirokov.nm@phystech.edu

caveats travel with the statement: the ghost scalar must exist at that epoch to hold the newborn throats open, and the merger’s scalar channel radiates negative energy whose cosmological consequence is genuinely open.

The paper is organised as follows. Sec. II sets out the model: the Einstein–phantom-scalar system, the drainhole, the binary superposition with its sign rule, and the declared seeds. Sec. III gives the numerical setup and the run matrix. The results then follow the campaign from one throat to two: the lone wormhole—fixed point, unstable mode, branching and collapse (Sec. IV); two throats at rest and the force between them (Sec. V); the head-on merger to a black hole (Sec. VI); the spiral merger and its capture boundary (Sec. VII); the gravitational waves of every channel—collapse, head-on, spiral, fly-by—gathered in one chapter with the vacuum twin (Sec. VIII); the numerical controls and systematics (Sec. IX). Sec. X confronts the rescaled waveforms with the LIGO–Virgo–KAGRA data products—template banks, ringdown, burst searches—and Sec. XI collects the findings. Every evolution runs on GPUs with GRTeclyn [39], the AMReX-based [40] port of GRChombo [41], continuing the campaign of Refs. [42, 43]. Throughout, $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so $\rho < 0$ where the field’s gradient dominates—the exotic stress that holds a throat open. With $V = 0$ this is exactly a ghost field. The sign flip lives entirely in the coupling to geometry: the scalar itself obeys the canonical wave equation $\square \phi = 0$, evolved first order in ϕ and $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$.

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi a^2 C^2 = a^2 + m^2$ (so $C \rightarrow 1/\sqrt{4\pi}$ as $m \rightarrow 0$), $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints

hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the *lapse* rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no grid resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$ with areal radius $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2}$; the production throat $a = 2$, $m = 1$ has $R_{\min} = 3.8895$ at $r_t = 1.618$. The point $r \rightarrow 0$ is the far universe’s spatial infinity compactified ($\chi \sim r^4$). The mass matters: at $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two of them do not attract at all.

C. The binary superposition

Two throats at separation d (one-body radii r_A , r_B) are superposed in excess form,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so $\phi \rightarrow 0$ at the boundary, which is free for a massless field. Momentum enters through one Bowen–York term [33] per throat,

$$\hat{A}_{ij} = \frac{3}{2r^2} [P_i n_j + P_j n_i - (\delta_{ij} - n_i n_j) P \cdot n], \quad (5)$$

converted by $\tilde{A}_{ij} = \chi^{3/2} \hat{A}_{ij}$, with $K = 0$. Since $\Pi = 0$ the matter momentum density vanishes and $\hat{A}_{ij}$ is flat-space divergence free, so the momentum constraint is satisfied *exactly* at $t = 0$; only the Hamiltonian constraint carries the superposition defect, $O(a^2/d^2) + O(m/d)$. The relative sign σ is the pair’s one discrete parameter: $\phi \rightarrow -\phi$ is exact for a single body, but a phantom mediator *repels* like scalar charges and *attracts* opposite ones, with $|F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 > 1$ always ($= 5$ for the production throat)—so like-signed pairs can never merge under their own gravity, and $\sigma = -1$ is the merger channel. A windowed one-body correction of the Helfer type [34] is implemented but off in production (Sec. IX A).

D. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi [1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (6)$$

which moves the throat’s areal radius by $\approx 2\varepsilon$ with the sign of ε , and a quadrupolar factor

$$\psi \rightarrow \psi [1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (7)$$

about the z axis through the throat. K_{ij} and Π are untouched, so the momentum constraint stays exact.

The spherical seed selects the branch; the quadrupolar seed has no $\ell = 0$ part—it gives a collapsing throat a quadrupole to shed—and with both on the cross term is $O(\varepsilon\varepsilon_2)$.

III. NUMERICAL SETUP

A. Evolution scheme

Evolutions use `GRTEClyn`: CCZ4 [35] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, $1 + \log$ slicing [36] started from the exact static lapse $\alpha = e^u$ of Eq. (3), a Gamma-driver shift [37, 38] with $\eta = 1$ started from zero, fourth-order stencils, RK4, and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor; the bracket in Δt that makes it necessary is a control (Sec. IX A).

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ with the system at its centre; the base grid is 128^3 ($\Delta x = 0.5$). Refinement is by moving nested boxes centred on the tracked throats, shrinking by a factor of two per level (half-width 4 per throat at the production finest level) and re-drawn every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms raise this to levels 4–7 where stated, and one control doubles the box to $L = 128$. Outer boundaries are Sommerfeld, backed by an absorbing sponge over $r = 24$ –32 that ramps extra dissipation up to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth into the layer; everything dynamical sits inside $r = 24$.

Each throat carries a compactified second infinity at its own centre, where $\chi \sim r^4$, so the evolved variables are flooded: $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant keeps χ unfloored down to 10^{-20} and applies the 10^{-8} floor only where $1/\chi$ is evaluated). Two further devices act only inside regions that are already trapped: an exponential damping of the scalar where the lapse has collapsed, and a smooth interior freeze that multiplies every right-hand side by $1 - W(r)$, with W a C^2 quintic step between an inner radius r_{full} and an outer r_{start} . All three are numerical devices, not physics; whether any of them is load-bearing is itself measured by twin runs (Secs. VIB and IX A).

C. Diagnostics

L^2 norms of the Hamiltonian and momentum constraints over the box are written continuously. Both throat centres are tracked, and about each centre the areal radius $R(r)$ is scanned on coordinate spheres; its minimum is the throat radius. On the same spheres the two null expansions $\theta_{\pm}$ are computed with the outward direction fixed by *increasing areal radius*: a

marginally outer trapped surface is an outermost zero of θ_+ with $\theta_- < 0$ (crossings at extrema of $R(r)$ are rejected), while a dynamic wormhole throat is marginally *anti*-trapped [32]. Masses of such surfaces are Misner–Sharp [31], $2M/R = 1 + \frac{1}{4}R^2\theta_+\theta_-$; the scan uses coordinate spheres rather than a deformable finder, and quoted masses carry that shape systematic. Radiation is read from Ψ_4 interpolated onto spheres at $R = 10, 14, 18$ (and 22 on the single-throat wave arms), decomposed in spin-weight -2 harmonics; all amplitudes are quoted as $r\Psi_4$. The in-code stream is cross-checked against an independent plotfile pipeline at the same radii. A per-cell autopsy reports the location and one-step history of any floating-point failure.

D. The campaign

The campaign is one-knob: every run differs from a named reference in a single setting, and every run has one line—its difference, its outcome, its caveats—in a registry shipped with the results pack (Sec. XII). Table I groups the 130 evolutions; the verdicts are the business of Secs. IV–IX. An archived shakedown stage (12 of the 130) fixed the gauge, dissipation and tagging choices of Secs. III A–III B and is not used for physics.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved with no declared perturbation, the exact drainhole holds its areal throat radius $R_{\star} = 3.8895$ to better than 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$; at level 4 the radius is still exact to four digits at $t = 30$. The static throat is a fixed point of the discrete system as well as of the continuum one. What ends the quiet phase is not a failure of the data: the departure time moves *later* by 5.6–8.7 units per halving of the cell, the signature of a truncation-noise seed of fixed growth rate being handed a smaller initial amplitude. Level 2 never reaches the branch at all—it dies of an origin failure at $t = 24.2$ with the throat still exact to 0.12%.

B. The unstable mode

The departure is a single exponential mode (Fig. 1(a,b)): plateau fits give e-fold times $\tau = 5.88$ (level 3) and 5.26 (level 4)—the same mode to 11%—at $T = \tau\alpha_{\text{th}}/R_{\star} = 0.87$ and 0.78 against the González–Guzmán–Sarbach band 0.68–0.76 for the massless limit [14]. The instability is the throat’s; only its seed is the grid’s, and the seed’s *sign* flips with resolution: level 3 collapses, level 4 inflates. This confirms, from exact massive-drainhole data, the unstable behaviour

TABLE I. The run matrix: 130 evolutions, grouped. Every group varies one knob against a named reference; per-run lines live in the shipped registry. Body parameters are $a = 2$, $m = 1$ ($R_{\min} = 3.89$), $\sigma = -1$, $d = 12$ and $L = 64$ unless stated. Left: the single throat, the pair and the head-on, with the controls. Right: the orbital family.

group	knob(s)	runs	Sec.	group	knob(s)	runs	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	9	IV	capture scan	$p = 0.20\text{--}0.45$	4	VII A
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; level 4	8	IV C	fly-by, $p = 0.45$	level 5 on the doubled box	2	VIII D
quadrupolar kicks	$\varepsilon_2 = 0.005\text{--}0.05$; kick-free arm	6	VIII A	orbital chain, $p = 0.12$	gauge; N ; levels; damping; χ floor	16	VII
two throats at rest	σ ; $a = 1\text{--}3$; $d = 12\text{--}18$	9	V	refinement ladder	levels 3–7; Δt ; damping	8	VII C
placement probes	$d = 6\text{--}48$, one step each	18	V C	interior freeze, $L=64$	skin radii; arming time; extraction	7	VIII C
head-on, $d = 8$	fill; levels 3/5; down-step; $\pm\varepsilon$	9	VI	wall probes	$p = 0.15\text{--}0.25$; levels; fill off	6	VII C
half-mass, $m = 0.5$	pair at $d = 6$; lone control	2	VI C	production spiral, $L=128$	three-stage chain; fill-window twin	5	VII B
vacuum BBH	same d , p , masses; no scalar	2	VIII E	Helper twins	correction; window; levels	6	IX A
constraint-solved data	GRTresna bridge	1	IX A	shakedown (archived)	gauge, dissipation, tagging	12	III A

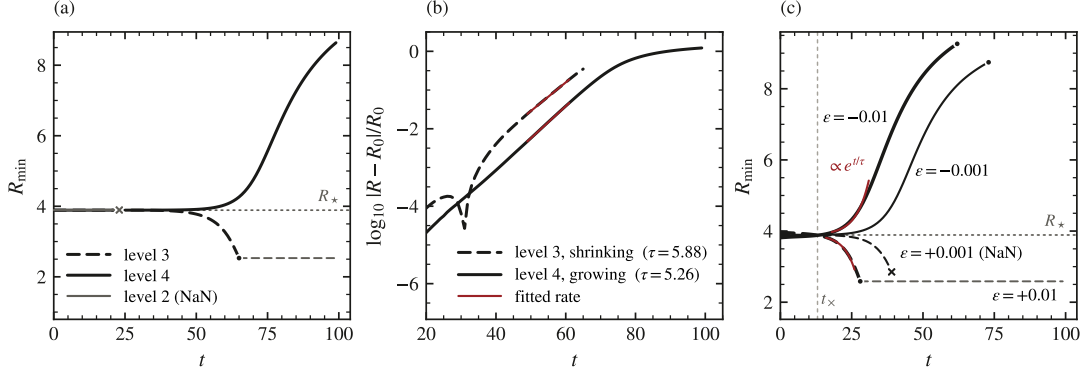


FIG. 1. The lone throat is unstable; unless the seed is declared, the grid picks the branch. (a) Minimal areal radius from exact static data at refinement levels 2–4: the exact solution is a fixed point (R_* , dotted), held until each grid’s own truncation noise excites the unstable mode—level 3 (dashed) collapses, level 4 (solid) inflates, level 2 (grey) dies of an origin failure first (cross). Dots mark where the *scan* loses the throat, not where the run ends; muted lines carry the last reading (the level-3 run continues to $t = 100$ behind a trapped surface). (b) The deviation is a single exponential mode with a level-dependent sign: plateau fits give $\tau = 5.88/5.26$ at levels 3/4, in the González–Guzmán–Sarbach band [14]. (c) Level 3 with the seed declared, Eq. (6): $\varepsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light); dashed = outward. The fate is opposite to the push, the twins cross at the ε -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives 8–14 units later depending on the marker. Burgundy: exponential fits $|R - R_*| \propto e^{t/\tau}$; cross: death at $t = 40.1$.

and the two-branch structure found for the massless Ellis–Bronnikov throat in Ref. [42], where ambient noise drove inflation and a compressive perturbation drove collapse: the branches are the wormhole’s, not an artefact of either code.

C. Branching: collapse versus inflation

A declared kick removes the grid’s vote (Fig. 1(c)). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses itself at the ε -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives later by a fixed delay—11–12 units read at matched horizon radius (8.2 by twin separation, 14 by first detection)—linear response riding the same unstable mode. The branching survives refinement: at level 4, the grid whose own truncation seed *inflates* the unkicked throat, $+10^{-2}$ still forms its horizon at $t = 11$ (areal 3.883 against level

3’s 3.88) and -10^{-2} still inflates, the twins crossing at $t = 12.99$ against 13.01. The kick also sets the clock: the collapse branch forms its horizon at $t = 11$ for $\varepsilon = +10^{-2}$ and $t = 25$ for $+10^{-3}$, against $t \simeq 61$ for the unkicked throat left to truncation noise—a driven branching an order of magnitude faster than the undeclared one. The dial has a ceiling: $\varepsilon = \pm 0.1$ is not a perturbation—both signs collapse and die ($t = 15.2/14.1$) with a Hamiltonian violation 3–5 $\times$ the 1% arms’ and growing—so the usable amplitude ends at 10^{-2} .

D. The collapse endpoint

The collapse branch ends in a black hole that eats its own support: behind the marginally trapped surface the areal radius falls $3.23 \rightarrow 2.37$ and the Misner–Sharp mass $1.62 \rightarrow 1.23$ over 40 units—phantom infall *removes* mass—with the origin lapse at 0.016 and no bounce by

$t = 100$. (The massless throat of Ref. [42] bounced at $t \approx 4M$ and shed its horizon by $18.5M$; the massive drainhole shows no bounce in 40 units.) The kicked $\varepsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon, and the pure-quadrupole arm gives the endpoint its full instrumentation (Fig. 2): a MOTS from $t = 33$ at areal radius 3.80 settling to 2.41 with $M_{\text{MS}} 1.90 \rightarrow 1.24$, while the radial shells watch $|K|$ grow inward and the lapse pinch to zero at the centre—the first single-throat collapse recorded by the in-code radial profile from beginning to end. The inflation branch forms no horizon at all: a growing anti-trapped shell, the throat +122% by $t = 100$, the compactified inner sheet deforming. Mass sets the stability margin from below: the lone $m = 0.5$ throat tracks its exact solution only to $t \approx 17$ at the 0.1% level (the $m = 1$ twin holds to $t \approx 35$) and dies at its origin at $t = 26.3$ with no horizon—the lighter throat is the less stable one.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat's scalar field turns the push into a pull (Fig. 3(a,b)). That one sign is the only gravity-driven route to a merger in this model, and every merging binary in this paper carries it. The measured pull-to-push ratio at $a = 2$ is 1.518 ± 0.021 over 15 slices against the predicted $3/2$, and the force follows the throat *width*—the scalar charge—not the mass, which is equal in all arms.

B. The force law

Four like-signed pairs, byte-identical but for the gap, give displacements at a common $t = 11.5$ of 0.4696/0.3699/0.2980/0.2438 for $d = 12/14/16/18$. Pure $1/d^2$ is excluded— $F d^2$ rises monotonically $67.6 \rightarrow 79.0$ across the ladder, a 15.4% spread—but $F \propto (d + \delta)^{-2}$ with $\delta \simeq 3.5$ fits every rung pair, and it was a blind prediction: δ fitted on $d = 12$ –16 alone predicts 0.243 at $d = 18$ where inverse-square predicts 0.209, and the measurement is 0.2438. Since δ is comparable to the measured mouth radius of the placed pair (4.29 at $a = 2$, $d = 12$; $R_\star = 3.89$ isolated, Sec. VC), the force is inverse-square in an *effective* separation exceeding the coordinate one by about a throat radius—the deviation lives in the distance label, not in the law. Across widths the push grows at every rung (0.147/0.283/0.416/0.615 for $a = 1/1.5/2/3$) with a measured exponent ≈ 1.36 and falling; the point-charge a^2 scaling is not confirmed.

C. Placement systematics

The superposition itself is on the ruler: freshly placed pairs read +1.8% ($d = 48$) to +20.5% ($d = 6$) wider than the isolated throat, falling as $d^{-1.16}$ (Fig. 3(c,d)), so every same- d comparison in this paper is made against that curve rather than against the lone throat. Read that way, the released head-on pair's mouths are genuinely squeezed, -0.6% at $t = 7$ growing to -4.0% by $t = 13$ —an interaction effect, not placement, since the lone throat is exact until $t = 35$.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest at $d = 8$ with opposite scalar signs, the pair closes to contact at $t \approx 20$ while both mouths are still open wormholes, and a common marginally outer-trapped surface encloses them from $t = 22$: the offline corrected scan finds it at $r = 3.17$, areal radius $R = 5.56$, Misner–Sharp mass 2.99, growing to $R = 5.71$ by $t = 24$ (Fig. 4). The per-mouth scans never find a surface at either mouth at any time—the horizon is born common or not at all. The formation is robust where the wall is not. The frozen-interior ensemble carrying $\varepsilon = \pm 10^{-2}$ on both throats—the amplitude that decided a *lone* throat's fate—finds the same surface at $t = 22.0$, with r_{mots} and M_{MS} within half a percent of the unkicked arm, and the $-\varepsilon$ member tracks that horizon to $t = 100$ within $\sim 3\%$ in radius and $\sim 1\%$ in mass; the same kick moves the level-3 death by more than a unit ($+\varepsilon$ dies at $t = 25.76$ against the scout's 26.91). The seed sets the throat's fate and the grid's lifetime; it does not set the pair's.

B. Through the wall to a black hole

Five units after the horizon forms, the level-3 scout dies at $t = 26.91$ with the classic anatomy— χ floored at the midpoint, $\max|K|$ running away to 18, a one-step overflow—but the failure is the grid's, not the space-time's: at $t = 27.05$, where the scout died, the level-5 arm reads a smooth, monotone midpoint (lapse 1.6×10^{-2} , $\chi = 1.2 \times 10^{-6}$, $\max|K| = 0.11$) and runs to $t = 100$ with the constraint norms *falling* the whole way. The remnant is a trapped ball with no throat left, losing mass to the phantom field it swallows ($M_{\text{MS}} 2.99 \rightarrow 2.17$, Fig. 4). The exterior is grid- and device-independent: the down-step arm lands on the level-5 constraints to 0.08%/1.5% and on its wave to 0.05–0.33% of peak over $t = 45$ –100, and the fill twins agree with the no-fill arms at the 0.01%-of-peak level.

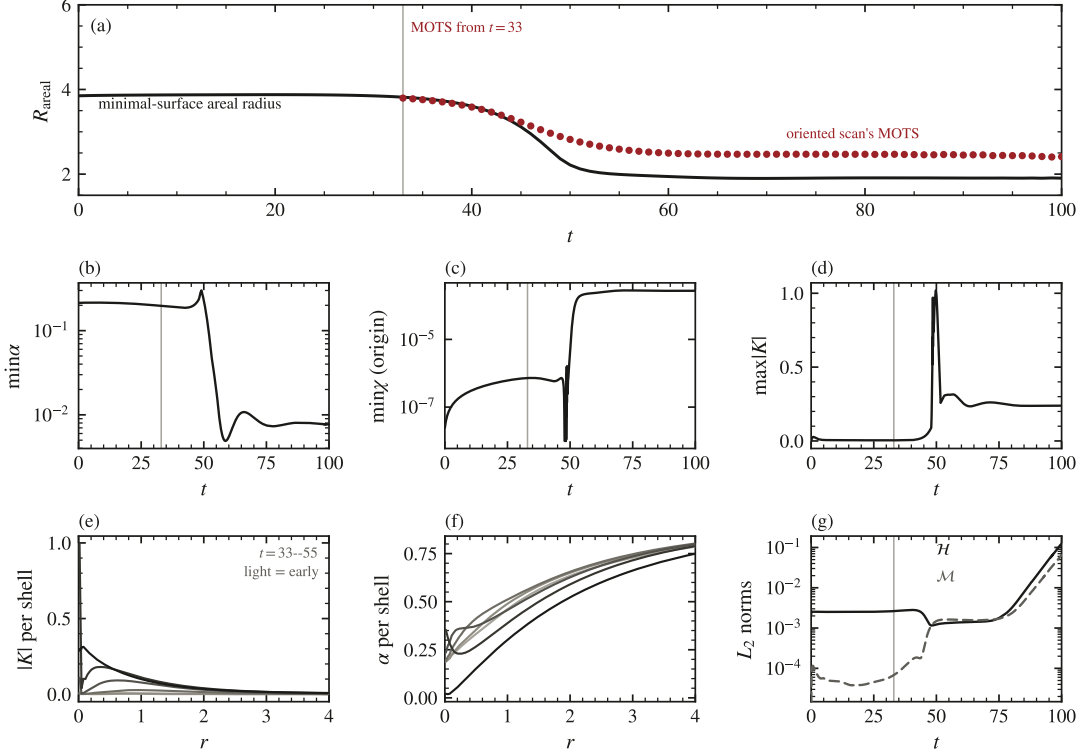


FIG. 2. The pure quadrupole’s collapse, fully instrumented (‘no kick’ arm: $\varepsilon_2 = 0.01$, radial kick zero). (a) Minimal-surface areal radius (ink) and the oriented scan’s MOTS (burgundy) from $t = 33$: R falls $3.80 \rightarrow 2.41$ and $M_{\text{MS}} 1.90 \rightarrow 1.24$ —phantom infall removes mass. (b)–(d) Global $\min \alpha$, $\min \chi$ (the origin monitor, the far sheet’s compactified infinity) and $\max |K|$; the violent phase is $t = 45$ – 55 . (e)–(f) The radial profile’s 128 shells over $t = 33$ – 55 (light = early): $|K|$ grows inward while α pinches to zero at the centre. (g) Constraint norms; the late rise from $t \gtrsim 75$ is this family’s known sphere-local floor growth, shared with the unkicked control.

C. The half-mass pair

Halving the throat mass does not avoid the horizon; it moves everything earlier. At $m = 0.5$, $d = 6$, the lapse falls from 0.44 to 0.036 over $t = 7.5$ – 13 , the tracker merges the mouths at $t = 11.1$, the offline scan confirms a common MOTS at $t = 12$ (areal $R = 3.83$, $M_{\text{MS}} = 2.16$), and the run dies at $t = 14.0$. The death belongs to the pair: the lone $m = 0.5$ control is exact to 0.02% through $t = 14$ and only fails, with no horizon of its own, at $t = 26.3$.

VII. THE SPIRAL MERGER

A. Capture and scattering

Dialling the tangential momentum at $d = 12$ splits the outcomes three ways: $p = 0.12$ and 0.15 fuse ($p = 0.15$ reproduces $p = 0.12$ ’s endgame signature—a plateau at pit separation 0.816 against 0.815, then the dive with the core lapse rising); $p = 0.20$ and 0.25 are captured into bound orbits; $p = 0.35$ and 0.45 escape. The capture boundary therefore sits between $p = 0.25$

and 0.35—50–70% of the circular value—and the fusing boundary between 0.15 and 0.20. The $p = 0.45$ fly-by is its own result: a bound pass at closest approach 3.95 after which *no horizon ever forms*—the corrected scan reads zero surfaces on the whole record—and both mouths swell $\times 7.8$ in areal radius, that expansion, not any wall, driving the late constraint growth ($\text{corr}(\log R, \log \|\mathcal{H}\|) = 0.92$). The swelling is the constituents’ own unstable mode with the companion driving it: the mouths’ areal-radius e-fold time collapses from 399 units while the pair is far ($t = 0$ – 20) to 52 through the approach and 32 after closest passage—the tidal acceleration the lifetime budget of Sec. XI A runs on.

B. Plunge and collapse

Every “spiral” in this campaign is a plunge, not an inspiral: with $M_{\text{ADM}} \simeq 1$ per throat, $d = 12$ is $d/M = 6$ —the innermost-stable-circular-orbit separation—and the throat tracks of the momentum scan complete only 0.16–0.23 of a revolution before closest approach ($p = 0.20$ – 0.35). Whether a wider pair could inspiral instead is answered, in the negative, by the lifetime budget of Sec. XI A. The production spiral ($p = 0.12$, $L = 128$) falls

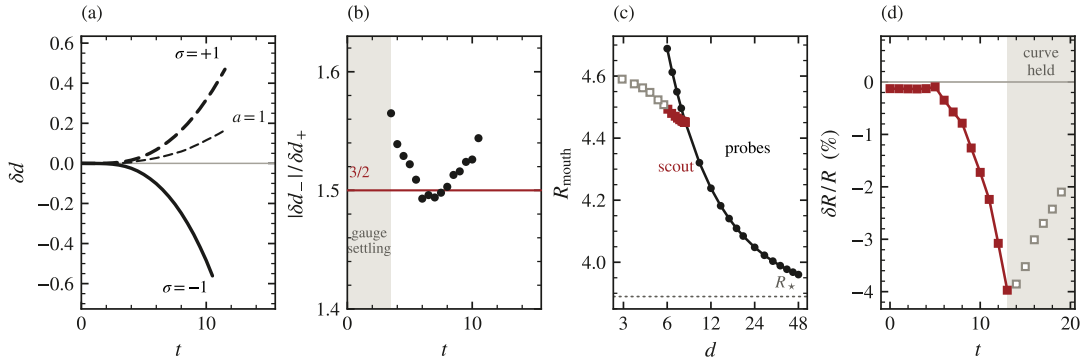


FIG. 3. Two throats at rest: the sign rule (a,b) and the ruler it is measured with (c,d). (a) Separation change of three $d = 12$ pairs one knob apart: the like pair ($\sigma = +1$, heavy dashed) pushes apart, flipping one field ($\sigma = -1$, solid) turns the push into a pull, and the narrow like pair ($a = 1$, light dashed) opens $2.8\times$ more slowly—the force follows the throat width, not the mass. Separations are inverse- χ -weighted centroids of the midplane χ pits, sub-cell where the tracker’s 0.03 grid quantum would swamp the early motion. (b) Pull over push: with $Q \equiv |F_\phi/F_{\text{grav}}| = 5$ (Sec. IIC), the prediction is $(Q+1)/(Q-1) = 3/2$ (burgundy); measured 1.518 ± 0.021 . The gauge-settling systematic (grey band) is common to both arms and cancels in the ratio. (c) Per-mouth areal radius of freshly placed pairs at $t = 0$ (ink circles): wider than the isolated throat (R_* , dotted) by an excess falling as $d^{-1.16}$ —the neighbour’s field on the ruler. Burgundy squares: the head-on scout’s mouths (Sec. VI) at the separation reached; open grey squares (here and in d) lie closer than the closest probe, where no calibration backs the comparison. (d) The scout against the curve, $\delta R/R = R_{\text{scout}}/R_{\text{place}} - 1$: the -0.13% plateau to $t = 5$ calibrates the tracker’s grid snap, and the throats’ own squeeze then grows to -4.0% by $t = 13$. In the shaded band the curve is held at its last point, so the magnitude is a lower bound; a boost can only raise a coordinate-sphere reading, so a reading below the curve is a real squeeze.

from $d = 11.9$ to a single merged pit and then collapses at the centre while the throat stays open (Fig. 5). The merger’s constraint transient at $t = 41.2$ is resolved, not resolution—the level-3 leg reproduces the level-5 norms to $3.5\%/0.7\%$ over the shared window—and everything after it is a clean gauge-and-curvature collapse: the lapse to 2.0×10^{-3} , χ to its floor at $t = 58.4$, $\max |K|$ up $25\times$ on a wall that lands on the throat. Unlike the head-on, the corrected scans find *no* marginally outer-trapped surface at any of $t = 55\text{--}60$, at either resolution, while the throat’s areal radius shrinks only 5.7% : at its death this remnant is still a wormhole.

C. The wall

No grid gets the spiral through its merger. Restarting the $p = 0.12$ run from the same $t = 50$ checkpoint at `max_level` 3–7, every arm dies on the same h_{11} NaN, and the gains saturate: $+1.03$, $+2.51$, $+0.53$, $+0.06$ units per doubling (Fig. 6). Sixteen times the resolution buys 4.13 units and never reaches the merger’s end, and a halved time step buys $+0.36$ on its own level-5 twin—most of the level $5 \rightarrow 6$ spatial gain—so the top of the ladder is saturation evidence, not a converged limit. The wall is insensitive to the matter device (the damped family dies on the same clock), to starting refined (`max_level` 5 from $t = 0$ dies at $t = 52.07$, indistinguishable from the restarted rung), and it moves only modestly when the whole problem is re-based: the production chain—base grid doubled, restarted three times earlier at $t = 36$ —buys 4.2 units

past the entire ladder and still dies, at $t = 60.445$. The interior freeze then carries the exterior to $t = 100$; it is a numerical device, engaged only after the burst is causally clear of the extraction spheres (Sec. VIIC).

VIII. GRAVITATIONAL WAVES

Every channel radiates, and the loudest is the one that never makes a horizon. At the innermost sphere the peak $|r\Psi_4|$ of the dominant mode reads: fly-by 4.1×10^{-2} , spiral 2.9×10^{-2} , head-on 2.3×10^{-2} , vacuum twin 1.1×10^{-2} , collapsing throat 2.9×10^{-3} (Fig. 7)—the scattering pass outradiates every merger, by $1.4\times$ over the spiral and $3.8\times$ over the equal-parameter vacuum binary. Each record propagates at c within the output cadence and falls as $1/r$ across its spheres, and rescaled to a detector the ordering survives: at each record’s own resolved spectral peak the strain amplitude spectral density reads $1.2 \times 10^{-21} \text{ Hz}^{-1/2}$ for the fly-by (193 Hz), 1.1×10^{-21} for the head-on and 9.3×10^{-22} for the spiral (both 406 Hz), 3.0×10^{-22} for the vacuum twin (450 Hz) and 1.5×10^{-22} for the lone throat (286 Hz) (Fig. 8). These are quoted at a stated frequency and not as maxima of the curve, because these bursts have no resolved strain maximum: after the $1/f^4$ weight every one of them still rises toward low frequency, so the maximum sits wherever the record’s own low-frequency corner stops it—a statement about record length rather than about the source.

The energy budget makes the same point without a detector: in the dominant multipole the fly-by radiates

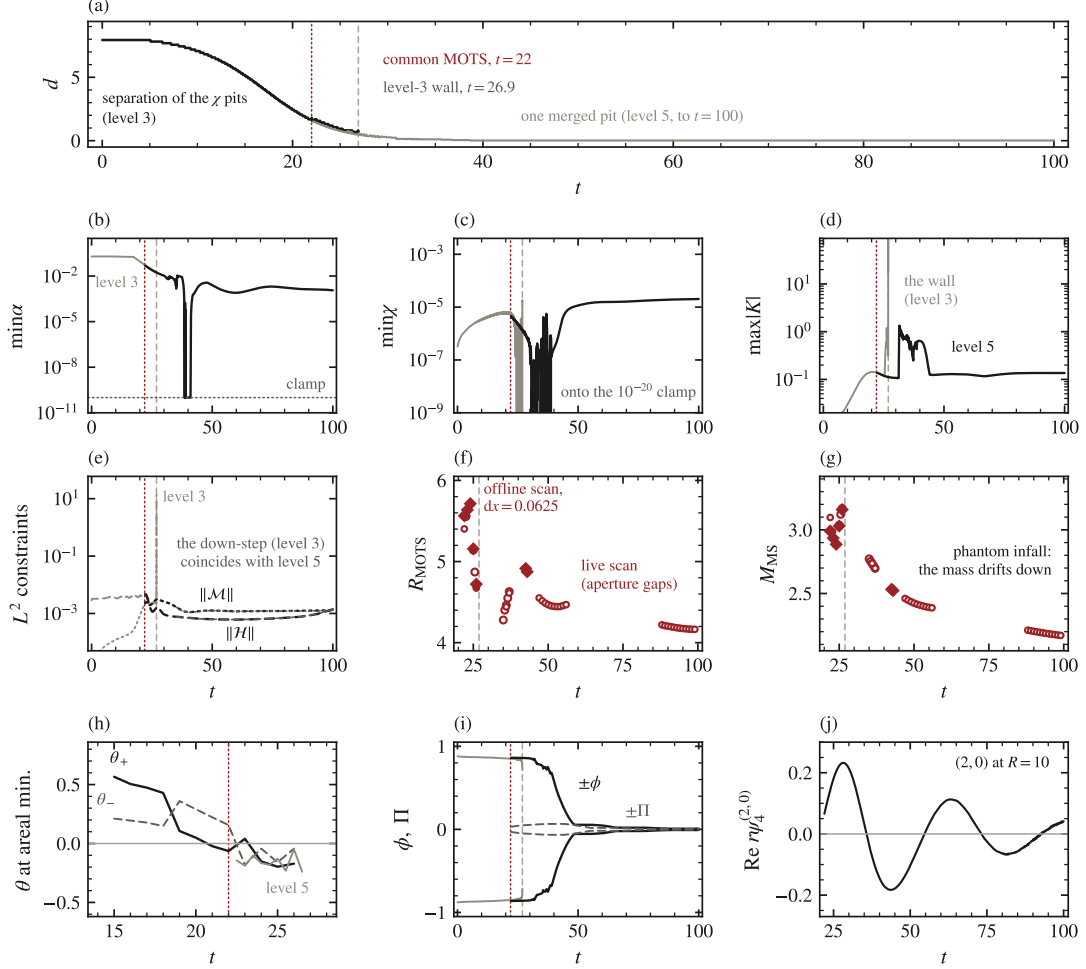


FIG. 4. The head-on merger makes a black hole ($d = 8$ from rest, opposite scalar signs); every horizon point is the orientation-corrected scan. (a) Separation of the χ pits: level 3 carries the approach from $d = 7.94$ (the tracker's grid snap on a true 8); the common MOTS forms at $t = 22$ (burgundy rule in every time panel), the scout dies at the level-3 wall at $t = 26.91$ (dashed rule), and the level-5 arm, restarted at $t = 22$ with every interior device off, carries the merged pit to $t = 100$ with no NaN. (b) The minimum lapse (level 5, ink) rides the 10^{-10} clamp over $t = 38.5\text{--}41.0$ and settles at 1.1×10^{-3} . (c) Minimum χ : excursions onto the 10^{-20} clamp leave the frame; level 5 returns and settles at 2.0×10^{-5} . (d) The wall is resolution: $\max |K|$ blows up to 18 at the level-3 death; level 5 never exceeds 1.34 and ends at 0.14. (e) The scout's $\|\mathcal{M}\|$ explodes at the wall (13.6); the level-5 norms fall $4.6 \times 10^{-3} \rightarrow 1.4 \times 10^{-3}$; the down-step (level 3 again, from the $t = 35$ checkpoint) lands on them to median 0.08%/1.5%. (f) The horizon: filled diamonds, offline scans at $dx = 0.0625$ (formation at $t = 22$, $r = 3.17$, $R = 5.56$; growth to 5.71 by $t = 24$; then 4.72, and 4.92/4.87 at $t = 42.5/43$); open circles, the live scan on three arms. Its gaps are aperture, not horizon loss—the shells reach $0.5d + 2.3$, so once the pits merge the surface at $r \simeq 2.7\text{--}3.3$ lies outside while every shell within reports trapped; where two arms see it at once they agree ($t = 36$: $R = 4.461$ against 4.441). (g) The Misner–Sharp mass drifts down, 2.99 at formation to 2.17 at $t = 99$ (-0.0066 per unit fitted on the down-step track): phantom infall removes mass. (h) Expansions at the areal minimum: θ_+ crosses zero between $t = 20$ and 21, both negative by $t = 24$; the per-mouth scans never find an own MOTS—the horizon is born common. (i) The field's extrema fall $\pm 0.86 \rightarrow \pm 0.008$ with $\|\Pi\| \leq 0.07$: the supporting scalar is swallowed and radiated, leaving vacuum around a black hole. (j) The remnant's $(2, 0)$ mode at $R = 10$ rings with period $\sim 31\text{--}38$, $\times 0.6\text{--}0.8$ per half-swing, as the mass in (g) declines.

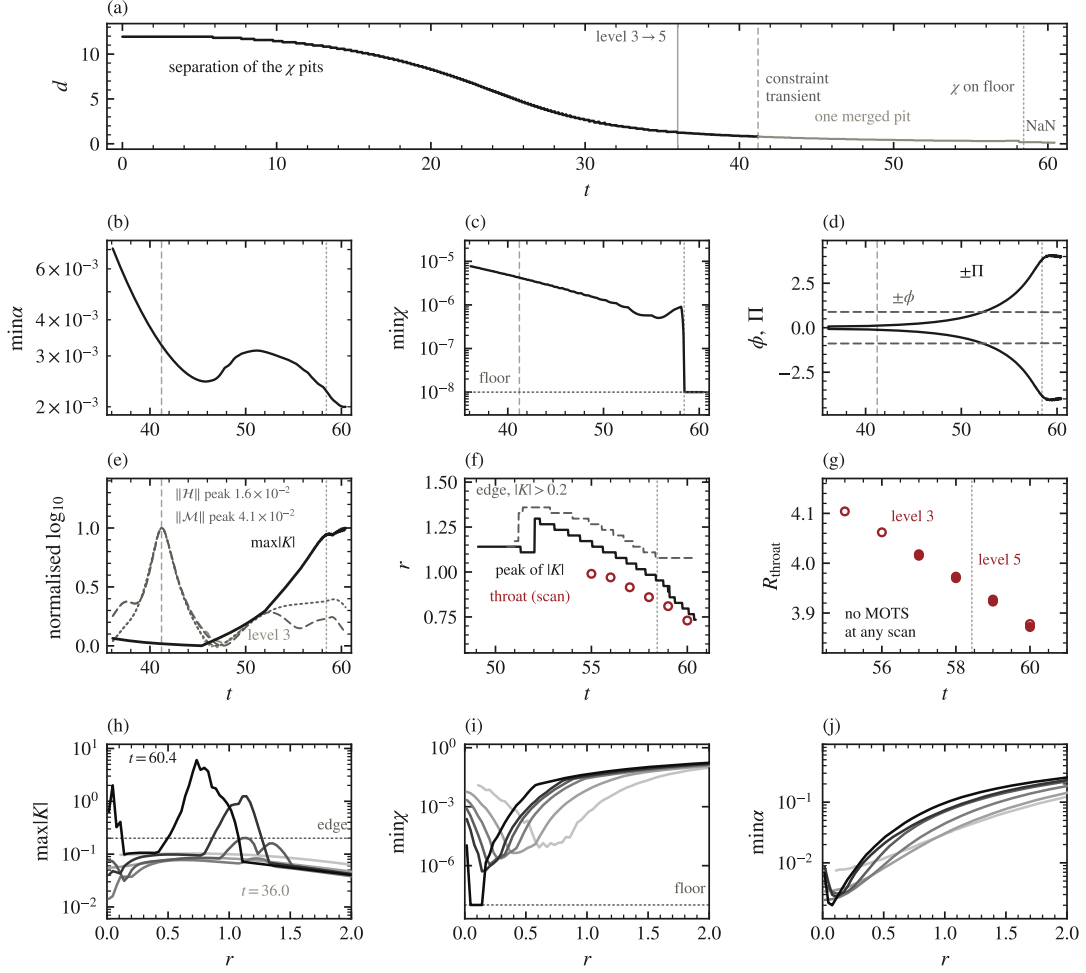


FIG. 5. The production spiral's collapse ($p = 0.12$, $d = 12$): level 5 from the restart at $t = 36$ to the death by NaN at $t = 60.44$, no panel involving the interior fill. Rules in every time panel: the handover at $t = 36$ (solid), the merger transient at $t = 41.2$ (dashed), the χ floor at $t = 58.43$ (dotted). (a) Separation of the two χ pits over the whole history—the level-3 leg carries the plunge from $d = 11.9$, after which the tracker follows one merged pit (pale). (b)–(d) $\min \alpha$ bottoms at 2.0×10^{-3} without reaching zero; $\min \chi$ decays exponentially onto its 10^{-8} floor at $r = 0.078$; the scalar holds ± 0.89 while Π saturates at ± 4.1 —the profile survives, the momentum diverges. (e) Growth compared, each curve $\log_{10}$ -scaled onto $[0, 1]$: the constraint norms peak at the merger and relax below where they started, while $\max |K|$ climbs only from $t = 49$, with $\text{corr}(\log \max |K|, \log \|\mathcal{H}\|) = -0.35$ after $t = 40$ —the death is not a constraint blow-up. Pale: the level-3 leg through the same transforms, median 3.5% ($\mathcal{H}$) and 0.7% ($\mathcal{M}$) over $t = 36$ –50. (f) Radius of the $|K|$ spike (steps, drawn from $t = 49.1$; earlier the profile is flat and an argmax is noise) and the $|K| > 0.2$ edge (dashed), widest at $r = 1.36$. Burgundy: the throat radius from the corrected scans—the spike rides just outside the throat and lands on it by the death. (g) Throat areal radius $4.10 \rightarrow 3.87$ over $t = 55$ –60, levels 3 and 5 agreeing to 0.2%, with *no* marginally outer-trapped surface at any scan. (The in-code θ_+ columns are omitted: their naive $+r$ orientation manufactures trapped verdicts near a throat.) (h)–(j) Radial anatomy at six equally spaced times, light ($t = 36.1$) to dark ($t = 60.4$): per-shell $\max |K|$ grows $0.08 \rightarrow 6.1$ on a spike ~ 0.25 wide migrating inward, while the χ and lapse pits stay at $r \lesssim 0.15$ —the gauge collapse and the curvature wall are different places.

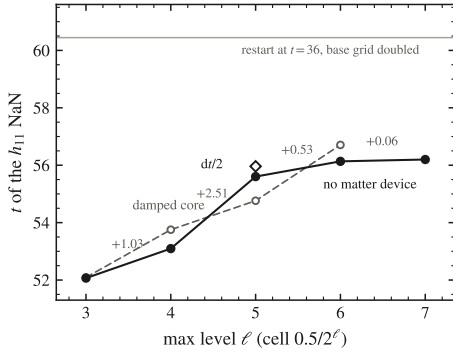


FIG. 6. The wall does not move: death time of the $p = 0.12$ spiral against refinement, every arm restarted from the same $t = 50$ checkpoint, every death an h_{11} NaN, no horizon at any rung. Filled: plain evolution, per-doubling gains marked. Open diamond: the level-5 arm at half the time step. Open circles: the core-damping family—a different treatment, dying on the same clock. Faint rule: the production chain (doubled base grid, restart at $t = 36$), 4.2 units past the whole ladder and still short of the merger’s end.

$7.4 \times 10^{-2} M$ and the spiral $2.2 \times 10^{-2} M$, against $2.6 \times 10^{-3} M$ for the equal-parameter vacuum binary— $29\times$ and $8.5\times$ more, and in the fly-by’s case more than the $4.84 \times 10^{-2} M$ a quasi-circular vacuum merger of the same mass radiates from infinity.

A. The collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor—by symmetry—while the same kick dressed with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, $35\times$ the no-quadrupole control, carrying $E_{\text{rad}} = 3.2 \times 10^{-5} M$. The burst arrives in radius order at $0.8\text{--}1.0 c$, carries the same $r\Psi_4$ to four spheres within 7.2%, and scales by 4.85 for a $5\times$ kick. Two further arms close the case (Fig. 9): halving the seed halves the wave (ratios 1.99/1.97/1.87 at $R = 10/14/18$), making the decade $\varepsilon_2 = 0.005\text{--}0.05$ linear to 1–16%; and a *pure* quadrupole—the same $\varepsilon_2 = 0.01$ with the radial kick removed—radiates the same wave to 2–3% while its throat still collapses (MOTS at $t = 33$, on the refinement level whose own truncation seed *inflates* the undeformed throat). The deformation, not the kick, is both the source of the wave and the selector of the fate. What is left rings at period $20.6 M$, 21% short of the Schwarzschild fundamental for the horizon mass; the decay time is not measurable on this record.

B. The head-on ringdown

The head-on’s signal is a bell with no chirp: the $(2, 0)$ breathing mode at $R = 10$ swings through $+0.23, -0.18$,

$+0.11, -0.07$ at $t = 28.2, 43.6, 63.0, 81.7$ —a period of $\sim 31\text{--}38$, lengthening, with the amplitude falling $\times 0.6\text{--}0.8$ per half-swing—and the same bell arrives ~ 4 units later per 4 units of extraction radius ($0.94 c$), outgoing across all three spheres. On the detector track it is the bell that slows from 486 to 342 Hz in Fig. 8(c).

C. The spiral’s carried burst

The spiral’s $(2, 2)$ burst peaks at $|r\Psi_4| = 2.9 \times 10^{-2}$ and folds onto one curve across $R = 20\text{--}44$ at $v/c = 0.95/1.00/1.00$. The burst is fill-free by causality: the interior freeze that carries the late record is armed at $t = 53$ with its skin at $r \geq 1.8$, so the earliest it can influence any extraction sphere is after the burst has passed it—and the head-on’s fill twins bound the device’s imprint at the 0.01%-of-peak level. The production arm now carries its own certificate (Fig. 10): a twin restarted from the same $t = 57$ checkpoint with only the fill window moved, $1.40/1.90 \rightarrow 1.25/1.75$, shifts the $(2, 2)$ waveform by at most 0.385% of its peak at $R = 20$ and 0.013% at $R = 44$ over the whole shared record $t = 57\text{--}100$ (overlap > 0.999999), with the residual fronts arriving on the fill’s own causal clock $t = 57 + (R - 1.9)$. The fill is a numerical device, not a source, and the carried burst survives having its container reshaped. What the burst is *not* is a black-hole birth cry: the corrected scans find no horizon anywhere on the record (Sec. VII B), so this is the wave of a merger that stalls as a wormhole.

D. The fly-by burst

The loudest signal in the campaign comes from the encounter that makes nothing: the $p = 0.45$ pass emits a single $(2, 2)$ arch of peak $|r\Psi_4| = 4.1 \times 10^{-2}$ with no ringdown—there is no remnant to ring—propagating at $v/c = 1.00$ across every sphere pair. The record is complete to $t = 100$ with no NaN and no horizon at any time; it is gated at $t = 70$. The mouths’ post-encounter expansion (areal radii $4.2 \rightarrow 33$ by $t = 97$, and $\text{corr}(\log R, \log \|\mathcal{H}\|) = 0.92$ —the swelling, not a wall, drives the late constraint growth) contaminates the innermost sphere, whose $|r\Psi_4|$ stops decaying at $t = 76.1$ and ends at twice the burst peak by $t = 100$. That turning point is where the contaminant has grown to *equal* the decaying burst, not where it arrives, so the gate is set ahead of it, 15.5 units past the $R = 20$ burst peak. The outer spheres are not a control here: in retarded time all four peak together at $t - R \simeq 34.5$, as radiation must.

E. The vacuum twin

The vacuum control—same d, p and masses, no scalar—does what black holes do: merges, climbs the point-mass chirp $299 \rightarrow 473$ Hz and peels off toward the

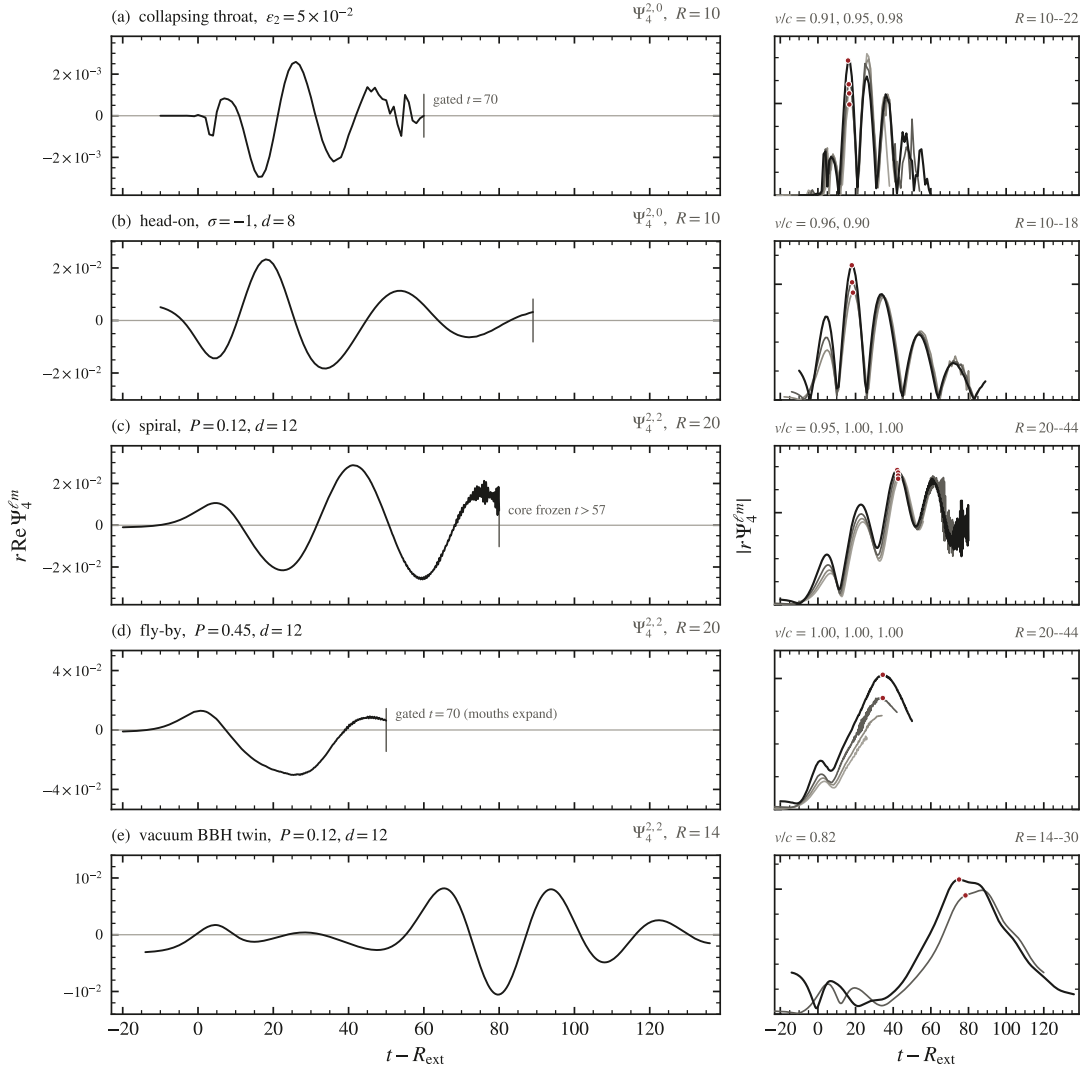


FIG. 7. The waveform gallery: every channel’s radiation, one row per scenario, one retarded clock. Left column: $r \text{Re} \Psi_4$ of the dominant mode at the innermost extraction sphere, each row on its own stated scale—the lone throat radiates an order of magnitude below the mergers, and the fly-by is the loudest channel of all. Half-height caps mark record ends: the collapsing throat’s stream is junk-gated at $t = 70$, the spiral record ends at $t = 60$ with the run still in flight, and the fly-by—complete to $t = 100$, no horizon ever forming—is gated at $t = 70$, ahead of the $t = 76.1$ trough where the mouths’ expansion (areal radii growing $\times 7.8$) has grown to equal the decaying burst at the $R = 20$ sphere. Right column: the envelope $|r \Psi_4^{\ell m}|$ at every extraction sphere on its own retarded time (ink to faint, inner to outer): a signal travelling at c and falling as $1/r$ folds onto one curve. Burgundy dots mark the correlated wavefront, and the quoted v/c is the lag of the whole complex waveform between neighbouring spheres over their common retarded window—peak-to-peak timing fails on flat-topped envelopes (the vacuum twin’s merger plateau is ~ 15 units wide at both spheres, and its peak times alone would read $v = 0.57c$ where the waveform lag reads $0.82c$). Measured speeds: throat $0.91/0.95/0.98$, head-on $0.96/0.90$, spiral $0.95/1.00/1.00$, fly-by $1.00/1.00/1.00$, vacuum twin 0.82 . Every channel propagates at c within the output cadence: radiation, not a gauge or constraint mode.

equal-mass Kerr mode, with its Ψ_4 peak at the textbook $fM = 0.066$. Against it the wormhole channels are loud: the fly-by’s peak $|r \Psi_4|$ is $3.8\times$ the twin’s and the spiral’s $2.7\times$. The twin also calibrates the wavefront method itself: its merger envelope is a ~ 15 -unit plateau at both spheres, where peak-to-peak timing reads a meaningless $0.57c$ and the full-waveform lag reads $0.82c$ —which is why every speed in this chapter is a correlated lag, never a peak time.

IX. CONTROLS AND SYSTEMATICS

A. Numerical systematics

Each numerical device carries a twin that measures it. The χ clamp is not load-bearing: twins at 10^{-8} against 10^{-20} re-converge to four digits after touching the floor, and the level-4 pair is byte-identical. The interior fill is not load-bearing for anything outside: head-on fill and

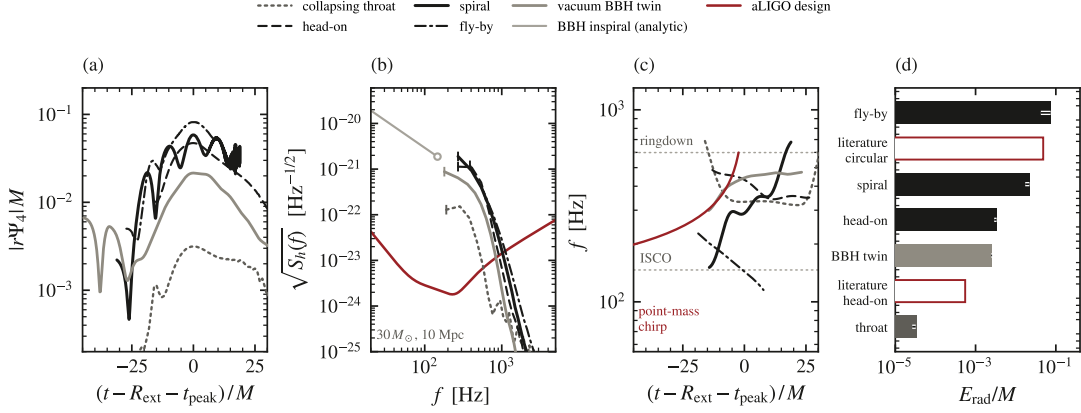


FIG. 8. Every source on a detector at one shared calibration: total mass $M = 30 M_\odot$, distance $D = 10 \text{ Mpc}$, each record first reduced to units of its own total ADM mass; strain scales as M/D , frequency as $1/M$. (a) Envelopes $|r\Psi_4|/M$ at the innermost sphere on a common merger clock ($\tau = 0$ at each peak): fly-by loudest (8.2×10^{-2}), then spiral, head-on, vacuum twin, lone throat (3.1×10^{-3}). (b) Strain amplitude spectral densities over the aLIGO design floor (burgundy), each curve drawn from its own low-frequency corner (one cycle per record, ticked), below which a record of length T cannot tell a wave from a drift. Read them as falling power laws (slopes -3.3 to -5.9) with no resolved maximum: any “peak” taken from this panel is the corner $1/T$; the values at the resolved Ψ_4 peaks are quoted in the text. Faint line: the Newtonian inspiral of the same binary, $|\dot{h}| \propto f^{-7/6}$ to its ISCO at 147 Hz—the ramp an astrophysical binary rides into the band where the simulated signal begins. (c) Instantaneous frequency over each record’s body against the point-mass chirp (burgundy), the ISCO (147 Hz) and the equal-mass Kerr ringdown (599 Hz): the vacuum twin climbs the chirp and peaks at $fM = 0.065$, the textbook merger value; no wormhole channel does (tracks in Sec. XC). (d) Energy radiated in the dominant multipole, ranked; the tick on each bar is the spread over extraction spheres, the error bar that catches near-zone content posing as a wave (fly-by 43%, the widest: the expansion of its open mouths is near-zone content the outer spheres see less of). Open burgundy bars: the published equal-mass vacuum results from rest at infinity—head-on infall 5.5×10^{-4} , quasi-circular merger 4.84×10^{-2} —which bracket any vacuum binary; the spiral and fly-by outradiate the vacuum twin by $8.5\times$ and $29\times$. Energies are band integrals of $|\int r\Psi_4 dt|^2$ over a common retarded window, doubled for the $\pm m$ pair; gravitational channel only—the scalar’s own energy is invisible to Ψ_4 .

no-fill arms agree to 0.01% of the wave peak. The diagnostics are provably non-invasive: two different binaries evolving the same spiral checkpoint produce byte-for-byte identical physics streams over 2384 shared output rows—which also certifies bit-reproducibility on this hardware. The time step is bracketed (dt multiplier 0.05 is a different solution from $t \approx 33$; 0.1 dies at $t = 16$; 0.02 is used throughout), grid independence after the head-on merger is 0.05–0.33% of peak across three restarted arms, and the placement systematic of Sec. VC is subtracted by construction. The Helfer superposition correction is rejected on measurement: better initial data, worse evolution. It trades the constraint defect at the throats for one in the fade zone between them, and the orbit then stalls at separation 4.7 instead of plunging—at levels 3, 4 and 5 alike, so converged rather than under-resolved, and worse (5.1) when the window is halved; on a fly-by, where nothing merges, the corrected arm’s Hamiltonian norm grows $3.2 \times 10^{-3} \rightarrow 1.6 \times 10^{-2}$ in seven units against a flat 2.2×10^{-3} for its plain twin. These arms do not die—they stall, and were stopped by hand—so production superposes plainly and accepts the initial mouth-size excess of Sec. VC. Constraint-solved data was attempted and is not used either: a *GRTresna* solve reproduces the drainhole once the outer condition on ψ_{reg} is Robin rather than unity (ψ error 4.9% $\rightarrow$ 0.7%), but the bridge that

carries the solved grid into the evolution loses the throat, dying at $t = 5.3$ —so every binary here is superposed, with the defect of Sec. IIC declared rather than removed. Sec. XIA shows that removing it would buy a fifth of an orbit. Matter damping shapes nothing it was hoped to shape. Finally, the late Ψ_4 floor growth of the single-throat family is sphere-local, not a boundary reflection: doubling the box leaves the onset at $t \simeq 41$, and in the spherical control—which cannot radiate a quadrupole—the $R = 14$ sphere climbs four orders of magnitude while the same run’s $R = 30$ sphere never leaves 1.1×10^{-3} . Every wave window in this paper is therefore gated per sphere, ahead of its own onset.

X. PATTERN MATCHING AGAINST DETECTOR DATA

The simulated waveforms are scale-free: with the throat mass M as the unit, $r\Psi_4$ rescales to detector strain by $h \propto M/D$ and $f \propto 1/M$, so one numerical waveform covers every candidate mass at once. This section confronts the rescaled (2, 2) and (2, 0) waveforms with the LIGO–Virgo–KAGRA data products on four fronts: their fitting factor against the template families the catalogues are built from [30], a matched-filter search of the

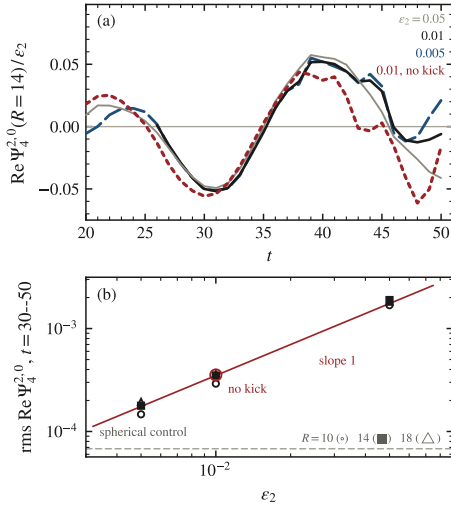


FIG. 9. One throat’s wave is linear in its deformation. (a) $\text{Re } \Psi_4^{2,0}$ at $R = 14$, each arm divided by its own seed: $\epsilon_2 = 0.005, 0.01, 0.05$ and the kick-free 0.01 arm collapse onto one curve. (b) rms amplitude over $t = 30-50$ against ϵ_2 at three spheres, with the slope-one line through the 0.01 arm: linear to 1–16% across a decade. The open circle is the pure quadrupole—no radial kick—matching the kicked arm to 2–3%; the dashed rule is the spherical control’s floor.

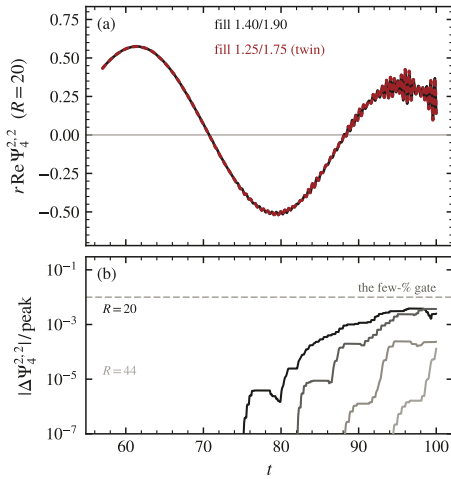


FIG. 10. Moving the fill window does not move the wave. (a) $r \text{Re } \Psi_4^{2,2}$ at $R = 20$ for the freeze arm (fill 1.40/1.90, ink) and its twin (1.25/1.75, burgundy dots), same $t = 57$ checkpoint, same binary: one bicolour curve. (b) The per-sphere envelope of the difference against the few-percent gate: at most 0.385% of peak at $R = 20$ and 0.013% at $R = 44$, each front arriving on the fill’s causal clock $t = 57 + (R - 1.9)$. The figure certifies the fill’s inertness, not a remnant ringdown.

open data itself [26], the ringdown, and the unmodelled searches.

One pipeline produces every number in this section, and it is validated before it is believed. The Ψ_4 records are integrated to strain by fixed-frequency integration [27], with the divisor clamped below a per-arm

corner—the lowest instantaneous frequency the record resolves, never below one cycle per record—without which the fly-by, whose frequency genuinely falls below its own $1/T$, is reconstructed a hundred times too loud. The residual pedestal $|h|(\text{ends})/|h|(\text{peak})$ is 0.02–0.07 for the collapsing throat and the vacuum twin, but 0.31–0.42 for the head-on, spiral and fly-by, whose low-frequency strain is therefore an upper bound and not a measurement; each template is filtered only from its own corner. The end-to-end test that can fail is the campaign’s vacuum binary-black-hole twin, a black-hole merger evolved with the same code, grid and extraction: through this pipeline it matches IMRPhenomD at 0.902–0.938 over 60–300 $M_\odot$ and recovers its total mass to 0–5% at 60–200 $M_\odot$, the missing few per cent accounted for by its eccentricity, its 75 M record and its finite extraction radius. Every fitting factor below is read against that 0.90–0.94 ceiling, never against unity.

A. Fitting factor against the binary-black-hole bank

Against a quasi-circular IMRPhenomD bank spanning 20–500 $M_\odot$ in total mass, mass ratios 1–4 and aligned effective spins ± 0.9 , at Advanced LIGO design sensitivity, the channels return fitting factors of 0.49–0.82: collapsing throat 0.54–0.79, head-on 0.49–0.75, spiral 0.54–0.80, fly-by 0.76–0.82. Taken alone those numbers would say a modelled search recovers 12–56% of the optimal detection volume. The control forbids that reading: the vacuum twin—an ordinary black-hole merger—scores 0.52–0.74 on the same bank, inside the wormhole range. What the full-bank fitting factor measures here is not the exoticism of the source but the *length of the record*: a bank template that spends seconds in band climbing its inspiral has most of its power with nothing to align against, and the recovered masses say the same (0.11–1.84 of the true value; the control’s own at 0.11–0.53). No claim that a wormhole merger would be missed by the modelled searches follows from these records—and none can, because the shortness is the physics rather than a truncation: this matter model cannot produce an inspiral at all (Sec. XI A), so the transient is the whole of the wormhole-phase signal, and what a modelled search would meet at late times is the chirp of whatever the throats convert into, outside these simulations.

What the records do settle is morphology at fixed duration. Cutting each bank template to the signal’s own window removes the duration deficit and leaves the shape: the channels return 0.82–0.98 against the control’s 0.94–0.95—at this duration, burst morphology alone does not separate a drainhole channel from a binary black hole. The discriminator that does is the frequency track, Sec. X C.

B. A matched-filter search of O3b open data

The rescaled waveforms were also run as an actual search of public Advanced LIGO strain [29]. The bank is the five channels times a mass ladder placed so that neighbouring rungs match at 0.97 against the live noise curve—127 templates, total mass the only bank dimension, because the records are scale-free—and the pipeline is the standard one [26, 28]: matched filter, Allen’s χ^2 bin test, re-weighted SNR, Hanford–Livingston coincidence, background from 5000 time slides. Its sensitivity is measured, not assumed: all 15 injections into conditioned O3b strain at optimal SNR 20 are recovered at 93–139% of optimal (median 103%), within 3 ms, with χ_r^2 near unity. The one weakness is the bank’s shortest template (5 ms, four χ^2 bins), where the veto demotes SNR 27.8 to 14.2—a cost in sensitivity at the lightest fly-by rungs, not in false triggers.

Searching 2.26 h of coincident CBC_CAT3 science time yields *no candidate at zero lag*. The background—13920 accidental coincidences over 471 days of slid livetime, loudest 60.4—caps the smallest reportable false-alarm rate at one per 471 days; its loudest event is an SNR 758 Hanford glitch that the veto re-weights to 60.4, raising the bar rather than entering the foreground.

The useful output of a null result is reach. At single-detector SNR 8 and optimal orientation, the median horizon over each channel’s mass ladder is 4.8 Gpc (spiral), 2.5 (fly-by), 1.8 (head-on), 1.3 (vacuum twin) and 0.15 (collapsing throat); divide by 2.26 for the sky-and-orientation average, and read the three pedestal-limited channels as upper bounds. The ordering is the energies’ ordering: the horizonless encounters are what a detector reaches furthest for.

C. Ringdown discriminators

Where the burst’s shape does not separate these channels, the way its frequency *moves* does. The reference is fixed by the control: the vacuum twin’s peak sits at $fM = 0.065$, the textbook equal-mass merger value, and its instantaneous frequency climbs $299 \rightarrow 473$ Hz along the Newtonian point-mass track at $30 M_\odot$ before peeling off at merger, heading toward the $\ell = m = 2$ Kerr quasinormal mode of the equal-mass remnant at $fM = 0.0885$ (599 Hz). That is what a binary black hole does: monotone rise, then a fixed ringdown frequency.

No drainhole channel does it (Fig. 8(c)). The head-on’s (2, 0) signal is a bell whose frequency *falls*, $486 \rightarrow 342$ Hz, over a sequence of swings of lengthening period (~ 31 – 38 in units of M) with the amplitude dropping only $\times 0.6$ – 0.8 per half-swing—a slowly decaying oscillation, not a ringdown, and it falls where a remnant’s would sit fixed at 599 Hz, a factor of 1.75 above where the bell ends. The fly-by falls further and faster, $227 \rightarrow 115$ Hz, as the pair separates; this one is expected, because no horizon ever

forms and there is no remnant to ring. The collapsing throat rings at a single frequency, 335 Hz at its peak ($fM = 0.050$, the 20.6 M period of Sec. VIII A), whose period is 21% *short* of the Schwarzschild fundamental for its Misner–Sharp mass $M_{\text{MS}} = 1.56$ (26 M)—it rings fast for its mass, at 0.55 of the equal-mass Kerr mode.

The practical statement is therefore narrow and defensible. A ringdown-only analysis that fits a damped sinusoid and asks whether its frequency and decay lie on the Kerr (M_f, a_f) track would reject every one of these channels—the head-on because its frequency drifts downward through the fit window instead of holding, the throat because it rings 27% fast for its mass, the fly-by because it is not a ringdown at all. Equivalently, a catalogue event whose post-merger frequency is measured to hold within a few per cent at the Kerr value excludes the head-on channel at that mass. Two limits travel with this. The spiral’s late record is transported by a frozen core from $t = 57$ (Sec. VIII C) and its frequency climbs $151 \rightarrow 679$ Hz without settling, so this campaign has not measured a spiral ringdown and quotes none. And the head-on’s bell is followed only over four swings on a record that ends at $t = 100$, so its decay constant is bounded, not fitted.

D. Burst searches and the stochastic background

An unmodelled search is the right instrument for these signals: a burst pipeline asks only for excess power, coherent between detectors, in a time–frequency region, so the missing inspiral—the feature that costs the modelled bank most of its match, and which this source class cannot supply—costs it nothing. The channels are short and loud in exactly its regime: 6–125 ms across the bank, the energy concentrated within a factor of three in frequency.

The clean, detector-independent deliverable promised in Sec. I is the energy per event in throat-mass units, and this campaign measures it in the dominant multipole, with the spread over extraction spheres as its error bar: fly-by $7.4 \times 10^{-2} M$ (43%), spiral $2.2 \times 10^{-2} M$ (21%), head-on $3.3 \times 10^{-3} M$ (13%), collapsing throat $3.2 \times 10^{-5} M$ (18%), against the equal-parameter vacuum binary’s $2.6 \times 10^{-3} M$ (5%). Since Ω_{GW} from a population scales as the product of the comoving event density and the energy released per event, these numbers convert any bound on Ω_{GW} —the observing-run limits, or the nucleosynthesis cap that holds outside every detector band [24, 25]—into a bound on the comoving abundance of primordial drainhole encounters, one that is a factor of 29 tighter for the horizonless fly-by than for an equal-mass black-hole binary of the same parameters. Three caveats travel with it, and none is removed by this work: the ghost scalar must exist at the epoch in question to hold the newborn throats open; the scalar channel radiates its own energy, of the opposite sign, which Ψ_4 does not see; and the fly-by’s 43% sphere spread is the widest in the campaign, because the expansion of its

open mouths is near-zone content that the outer spheres see less of.

XI. DISCUSSION

The four fates promised in Sec. I are all realised, each with a measured selection rule: the relative scalar sign separates scattering from merger (Sec. V), the seed’s sign separates collapse from inflation (Sec. IV), and the orbital momentum separates fusing, capture and escape at 50–70% of circular (Sec. VII). What a foam-born population radiates is bursts—the energies of Sec. XD convert any Ω_{GW} bound into an abundance bound—and the budget below shows why no member of the population chirps.

A. No wormhole inspiral

A drainhole binary cannot radiate an inspiral, and the obstruction is a logarithm. Three independent measurements fix the e-fold time of the mouths’ unstable mode at $\tau \simeq 4.4$ code units: an exponential fit to the fly-by’s mouth growth (back-extrapolated seed 7×10^{-4} , the superposition defect) gives 4.39; the eight-unit shift in departure time between refinement levels 3 and 4, divided by $\ln 16$, the fourth-order truncation ratio, gives ~ 2.9 ; and the 11–12-unit delay per decade of declared kick (Sec. IV C) gives ~ 5.0 . In proper time at the minimal surface, where the static lapse is $\alpha_{\text{th}} = e^{u(r_{\text{t}})} = 0.575$, that is 1.7–2.9, i.e. $T = \tau \alpha_{\text{th}} / R_{\star} = 0.43$ – 0.74 —bracketing the González–Guzmán–Sarbach band of Sec. IV B [14], now timed in a 3+1 evolution.

The budget follows. A quasi-circular pair at separation d needs $t_{\text{insp}} = \frac{5}{256}(d/M)^4 M/\eta$ to merge by radiation reaction (M the total mass, $\eta = 1/4$): 202 units from the innermost stable circular orbit $d/M = 6$ and 640 units from $d/M = 8$ —1.1 and 2.25 orbits, counting orbits at the initial separation ($T_{\text{orb}} = 2\pi\sqrt{d^3/M}$), which understates the cycles actually completed and so states the budget conservatively. The instability is thus handed $t_{\text{insp}}/\tau = 46$ e-folds at the ISCO and 146 for a two-orbit inspiral, so the mouths arrive at merger unchanged only if the initial perturbation is below $e^{-46} \simeq 10^{-20}$, respectively 10^{-63} . Turned around, the lifetime is $\tau \ln(1/\varepsilon)$: seeds of 10^{-6} , 10^{-10} , 10^{-15} buy 0.33, 0.55, 0.82 of one orbit—not one, for any nameable seed. The asymmetry is the whole argument: the orbit’s demand grows as $(d/M)^4$ while the throat’s endurance grows as $\ln(1/\varepsilon)$, so no finer grid, quieter data or better solver closes the gap—constraint-solved initial data, cutting the seed from the superposition defect to the truncation floor, is worth $\tau \ln(2 \times 10^3) \simeq 33$ units, a fifth of one orbit. A companion only sharpens the verdict: the fly-by’s tidal field drives the mode, collapsing the mouths’ e-fold from 399 units to 32 as the separation closes (Sec. VII A).

The wormhole-specific signal is therefore a short transient: within a few tens of M each mouth has collapsed—

forming a horizon at $t = 11$ and 25 for kicks of 10^{-2} and 10^{-3} —or inflated, measured to $\times 7.8$ in the fly-by, and the plunge, burst and conversion of Secs. VI–VIII are the whole of what this source class emits *as wormholes*. Three qualifications are mandatory. The statement covers the matter model the González–Guzmán–Sarbach theorem covers, a minimally coupled ghost scalar; throats held open by rotation, additional fields, thin shells or higher-curvature terms are untouched by it. The system converts rather than falls silent, so “no inspiral” means no *wormhole* inspiral: any long chirp belongs to the objects the throats become. And the seeds measured here are numerical—superposition defect and truncation noise—which the argument tolerates only because its dependence on the seed is logarithmic; no physical environment supplies a perturbation below 10^{-20} .

B. What is not established

Collected here so that no claim above has to carry its own retraction.

- 1. No horizon is measured for the spiral merger.** No trapped surface exists anywhere on its measured record, and the curvature wall that kills every grid rides the still-open throat; a $16\times$ refinement ladder saturates without reaching the merger’s end (Sec. VII).
- 2. No spiral ringdown is measured.** The late record is transported by a core frozen from $t = 57$, so its swings are propagation, not a remnant ringing; the instantaneous frequency climbs $151 \rightarrow 679$ Hz without settling (Sec. X C).
- 3. The converted pair’s own signal is not simulated.** After the wormhole transient the constituents are black holes or inflating runaways (Sec. XI A); the chirp of such a converted binary—what a modelled search would meet at late times—is outside these records, which bound only the wormhole-phase morphology (Sec. X A).
- 4. The merging spiral’s mouth sizes are unmeasured.** The production spiral carried no per-mouth areal-radius stream, so how deformed the throats already were when the burst was emitted is unmeasured; the fly-by, identical data at the same separation, is $+56\%$ in areal radius by $t = 40$, and the spiral’s burst is emitted near $t = 42$. Quantifying it needs the radius instrumented on a re-run.
- 5. The low-frequency strain of three channels is an upper bound.** Head-on, spiral and fly-by reconstruct with 31–42% of their peak strain still present at the record’s ends, so their sub-corner content is integration pedestal rather than measurement (Sec. X); the collapsing throat and the vacuum control, at 2–7%, do not have this problem.

6. **The late constraint growth on both single-throat branches from $t \approx 75$ is unexplained,** as is the grid-scale wobble that appears on the outermost head-on sphere after $t \approx 80$ (Sec. IX).

XII. REPRODUCIBILITY

Every evolution in Table I is one line in a registry shipped with the results pack—its single knob against a named reference, its outcome, its caveats—and every figure in this paper is regenerated by a script in the same pack from the archived diagnostic streams. The evolutions are bit-reproducible on the production hardware: two builds differing only in a passive diagnostic reproduce 2384 shared output rows byte for byte (Sec. IX). The code is `GRTEclyn` [39]; the drainhole initial data, the seeds of Sec. IID, the orientation-corrected surface scan and the parameter file of every run accompany the registry.

XIII. COMPUTATIONAL COST

Every evolution ran on one NVIDIA H100 80 GB device of a four-H100 node with two 32-core Xeon Platinum 8462Y+ processors—up to four arms concurrently while the cards were free—with `GRTEclyn` built CUDA-native against AMReX 26.02. At $\Delta t = 0.02 \Delta x$ a production arm ($L = 64, 128^3$ base, three refinement levels) advances 12–18 code units per hour, so a $t = 100$ evolution costs 6–8 h. Depth is the expense: the level-5 arms run at 2–4 units per hour—18.6 h for the head-on’s $t = 22 \rightarrow 100$, 46.7 h for the fly-by’s full $t = 0 \rightarrow 100$ on the doubled box, which peaks at 58 of the 80 GB with 26-GB checkpoints against a rolling keep-list—and the production spiral’s three-stage chain (Sec. VII C) costs 23.5 h end to end. The packed campaign totals 832 GPU-hours of evolution, recomputed from each run’s own logged speed and time span by a script shipped with the results pack.

ACKNOWLEDGMENTS

This work was supported by Gravity Frontiers (<https://www.gravityfrontiers.org/en>), which funded the underlying research and the development of the numerical-simulation software. The author extends sincere gratitude to Ilya Nachevsky for generously providing the high-performance computing resources essential to this work, and acknowledges the GRTL Collaboration for `GRChombo`, `GRTresna`, and `GRTEclyn`.

-
- [1] J. A. Wheeler, “Geons,” *Phys. Rev.* **97**, 511 (1955).
 - [2] J. A. Wheeler, “On the nature of quantum geometrodynamics,” *Ann. Phys. (N.Y.)* **2**, 604 (1957).
 - [3] C. W. Misner and J. A. Wheeler, “Classical physics as geometry,” *Ann. Phys. (N.Y.)* **2**, 525 (1957).
 - [4] T. A. Roman, “Inflating Lorentzian wormholes,” *Phys. Rev. D* **47**, 1370 (1993).
 - [5] J. Garriga, A. Vilenkin, and J. Zhang, “Black holes and the multiverse,” *J. Cosmol. Astropart. Phys.* **02**, 064 (2016).
 - [6] H. Deng, J. Garriga, and A. Vilenkin, “Primordial black hole and wormhole formation by domain walls,” *J. Cosmol. Astropart. Phys.* **04**, 050 (2017).
 - [7] A. A. Kirillov and E. P. Savelova, “Dark matter from a gas of wormholes,” *Phys. Lett. B* **660**, 93 (2008).
 - [8] A. A. Kirillov and E. P. Savelova, “Density perturbations in a gas of wormholes,” *Mon. Not. R. Astron. Soc.* **412**, 1710 (2011).
 - [9] H. G. Ellis, “Ether flow through a drainhole: A particle model in general relativity,” *J. Math. Phys.* **14**, 104 (1973).
 - [10] K. A. Bronnikov, “Scalar-tensor theory and scalar charge,” *Acta Phys. Polon. B* **4**, 251 (1973).
 - [11] M. S. Morris and K. S. Thorne, “Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity,” *Am. J. Phys.* **56**, 395 (1988).
 - [12] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).
 - [13] H. Shinkai and S. A. Hayward, “Fate of the first traversible wormhole: Black-hole collapse or inflationary expansion,” *Phys. Rev. D* **66**, 044005 (2002).
 - [14] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution,” *Class. Quantum Grav.* **26**, 015011 (2009); see also **26**, 015010 (2009).
 - [15] R. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).
 - [16] F. J. Tipler, “Singularities and causality violation,” *Ann. Phys. (N.Y.)* **108**, 1 (1977).
 - [17] S. W. Hawking, “Wormholes in spacetime,” *Phys. Rev. D* **37**, 904 (1988).
 - [18] Ya. B. Zel’dovich and I. D. Novikov, “The hypothesis of cores retarded during expansion and the hot cosmological model,” *Sov. Astron.* **10**, 602 (1967).
 - [19] B. J. Carr and S. W. Hawking, “Black holes in the early Universe,” *Mon. Not. R. Astron. Soc.* **168**, 399 (1974).
 - [20] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, “Constraints on primordial black holes,” *Rep. Prog. Phys.* **84**, 116902 (2021).
 - [21] C. Caprini and D. G. Figueroa, “Cosmological backgrounds of gravitational waves,” *Class. Quantum Grav.* **35**, 163001 (2018).

- [22] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr data set: Evidence for a gravitational-wave background,” *Astrophys. J. Lett.* **951**, L8 (2023).
- [23] P. Amaro-Seoane *et al.*, “Laser Interferometer Space Antenna,” arXiv:1702.00786 [astro-ph.IM] (2017).
- [24] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo’s third observing run,” *Phys. Rev. D* **104**, 022004 (2021).
- [25] L. Pagano, L. Salvati, and A. Melchiorri, “New constraints on primordial gravitational waves from Planck 2015,” *Phys. Lett. B* **760**, 823 (2016).
- [26] B. Allen, W. G. Anderson, P. R. Brady, D. A. Brown, and J. D. E. Creighton, “FINDCHIRP: An algorithm for detection of gravitational waves from inspiraling compact binaries,” *Phys. Rev. D* **85**, 122006 (2012).
- [27] C. Reisswig and D. Pollney, “Notes on the integration of numerical relativity waveforms,” *Class. Quantum Grav.* **28**, 195015 (2011).
- [28] S. A. Usman *et al.*, “The PyCBC search for gravitational waves from compact binary coalescence,” *Class. Quantum Grav.* **33**, 215004 (2016).
- [29] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Open data from the third observing run of LIGO, Virgo, KAGRA and GEO,” *Astrophys. J. Suppl.* **267**, 29 (2023).
- [30] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run,” *Phys. Rev. X* **13**, 041039 (2023).
- [31] C. W. Misner and D. H. Sharp, “Relativistic equations for adiabatic, spherically symmetric gravitational collapse,” *Phys. Rev.* **136**, B571 (1964).
- [32] D. Hochberg and M. Visser, “Dynamic wormholes, anti-trapped surfaces, and energy conditions,” *Phys. Rev. D* **58**, 044021 (1998).
- [33] J. M. Bowen and J. W. York, “Time-asymmetric initial data for black holes and black-hole collisions,” *Phys. Rev. D* **21**, 2047 (1980).
- [34] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim, “Malaise and remedy of binary boson-star initial data,” *Class. Quantum Grav.* **39**, 074001 (2022).
- [35] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating solutions,” *Phys. Rev. D* **85**, 064040 (2012).
- [36] C. Bona, J. Massó, E. Seidel, and J. Stela, “New formalism for numerical relativity,” *Phys. Rev. Lett.* **75**, 600 (1995).
- [37] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, “Accurate evolutions of orbiting black-hole binaries without excision,” *Phys. Rev. Lett.* **96**, 111101 (2006).
- [38] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, “Gravitational-wave extraction from an inspiraling configuration of merging black holes,” *Phys. Rev. Lett.* **96**, 111102 (2006).
- [39] GRTL Collaboration, **GRTEclyn**, <https://github.com/GRTLCollaboration/GRTEclyn>.
- [40] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).
- [41] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).
- [42] N. M. Shirokov, “Wormhole dynamics: nonlinear collapse and gravitational-wave emission,” arXiv:2604.00071 [gr-qc] (2026).
- [43] N. M. Shirokov, “The Bondi dipole in full numerical relativity: a self-accelerating positive–negative mass binary,” submitted (2026).